

TD1- Business Intelligence

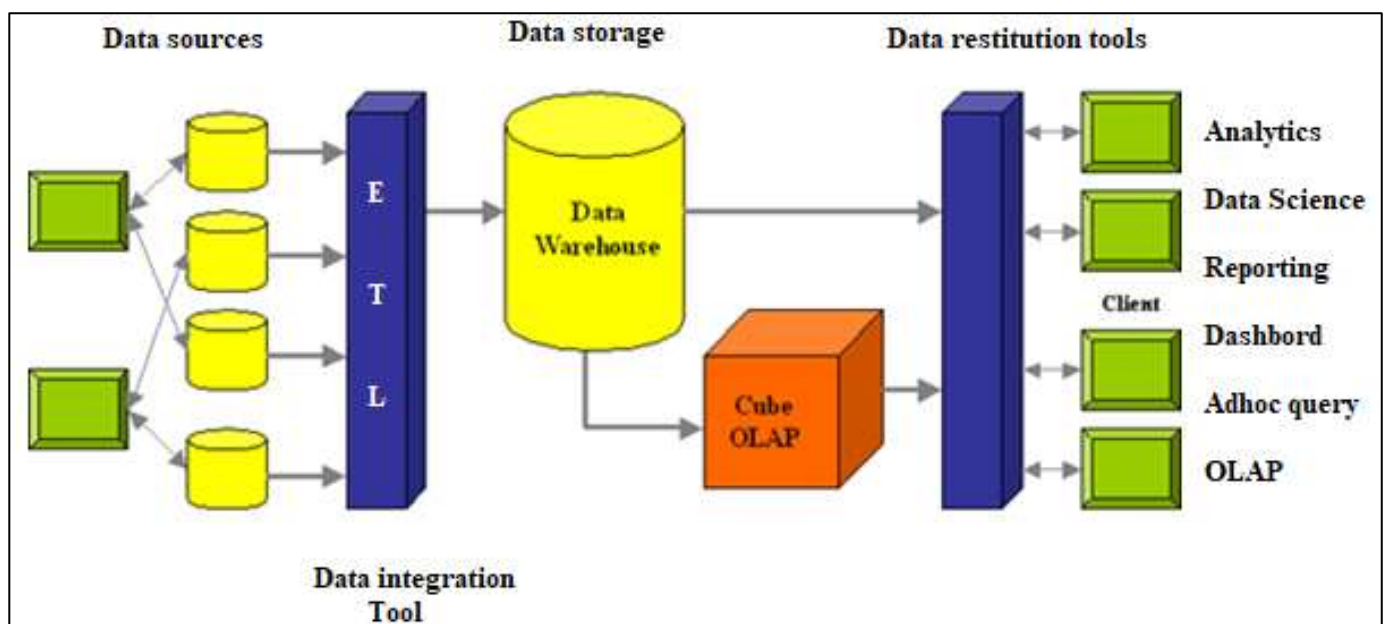
Educational objective: The objective of this tutorial is to discuss with the students the concepts already acquired relating to companies, the information system and decision support, to then make the link and introduce the concept of business intelligence.

1. Reminder Concepts:

- The company (definition, main objectives, role of the management function - planning, CTRL, organization, problem solving,...)
- Reminder of the importance of information systems for organizations, particularly businesses, in operational, technical, and strategic management and in decision support.
- Reminder of the types of decisions, and discussion of the information that must be provided for decision-making in the three levels of management.
- The Information System → is the IS that will provide the company with information, it will help the company in its management.

2. Introduction of BI and the components of a BI system in an interactive approach:

- « Business intelligence » (BI) refers to a broad range of procedures, approaches, technologies, and equipment used to extract useful information from unprocessed data and enable organizations to make well informed decisions. In order to improve business performance and competitiveness.
- In the Gartner IT Glossary, « BI is an umbrella term that includes the applications, infrastructure and tools, and best practices that enable access to and analysis of information to improve and optimize decisions and performance »
- BI system : It is a decision support information system that enables managers to analyse and interpret complex data about the company and its environment using simple tools..



- **The key components of Business Intelligence include¹:**
 - **Data sources :** Gathering data from disparate sources, both internal (e.g., transactional systems, CRM databases, ERP databases) and external (e.g., market research, social media),
 - **Data integration :** raw data are gathered (extracted), transformed and integrated into a centralised repository for analysis.
 - **Data Warehousing and Storage:** Storing structured and unstructured data in a centralized data warehouse, optimizing data accessibility and scalability for analytics.
 - **Analytics and Reporting:** Employing various analytical techniques, such as descriptive, diagnostic, predictive, and prescriptive analytics, to derive insights and generate actionable reports and dashboards ; interactive data visualization tools are used to convey complex insights in a meaningful and accessible manner, facilitating data driven decision-making at all levels of the organization.

- **Significance of Business Intelligence System :** the growing significance of Business Intelligence lies in its ability to empower organizations with timely, accurate, and actionable insights, enabling them to:
 - **Enhance Decision-Making:**
 - **Improve Operational Efficiency:** BI helps identifying process inefficiencies and bottlenecks, BI helps streamline operations, enhance productivity, and reduce costs.
 - **Gain Competitive Advantage:** BI enables organizations to identify market trends, customer preferences, and emerging opportunities, fostering innovation and market responsiveness.
 - **Enhance Customer Experience:** By analysing customer behaviour and preferences, BI facilitates personalized marketing strategies, customer segmentation, and targeted engagement.

¹ Muhammad Asif Khan, Introduction to BI : Concepts, Technologies and Applications.